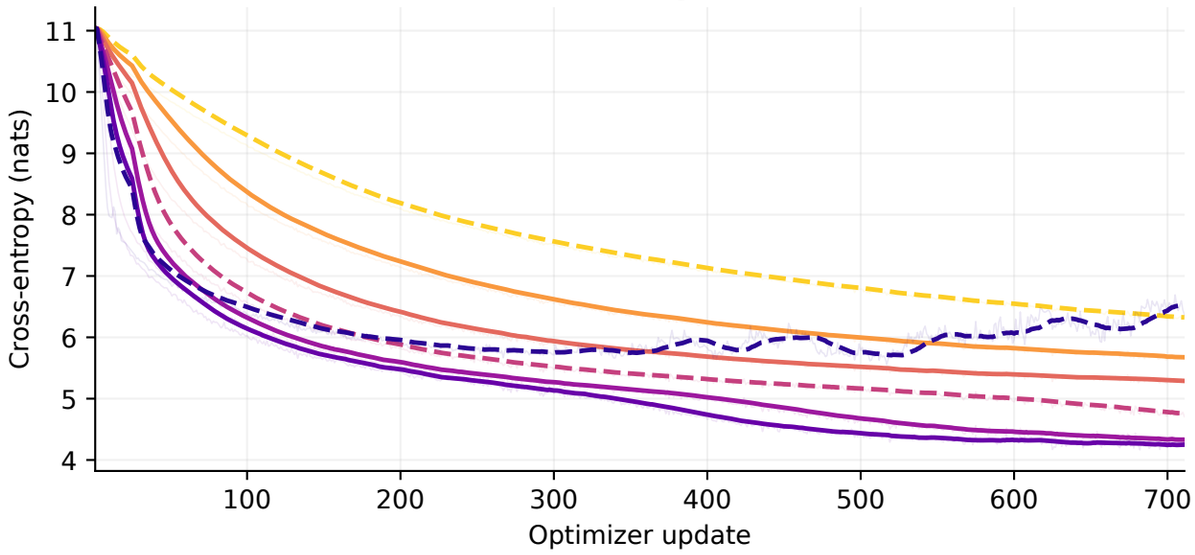
Pythia-14M: loss, raw cumulative geometry, alignment and vector norms

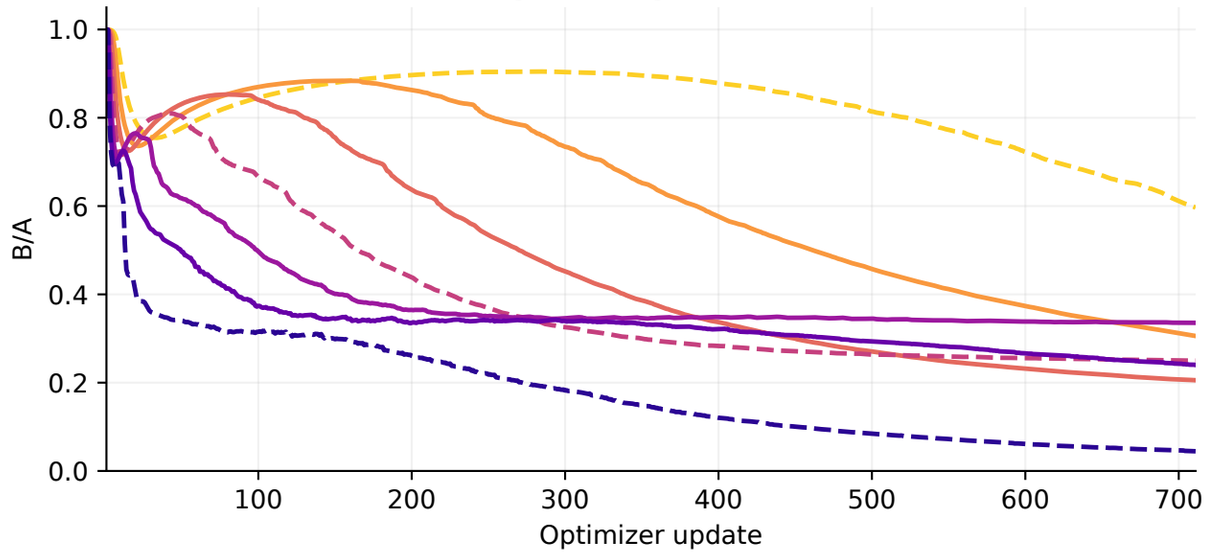
Peak LR: light yellow to dark purple as LR increases; S = Spark, R = RTX4090
Dashed = Spark; solid = RTX4090

0.0001 (S) · 0.000215443 (R) · 0.000464159 (R) · 0.001 (S) · 0.00215443 (R) · 0.00464159 (R) · 0.01 (S)

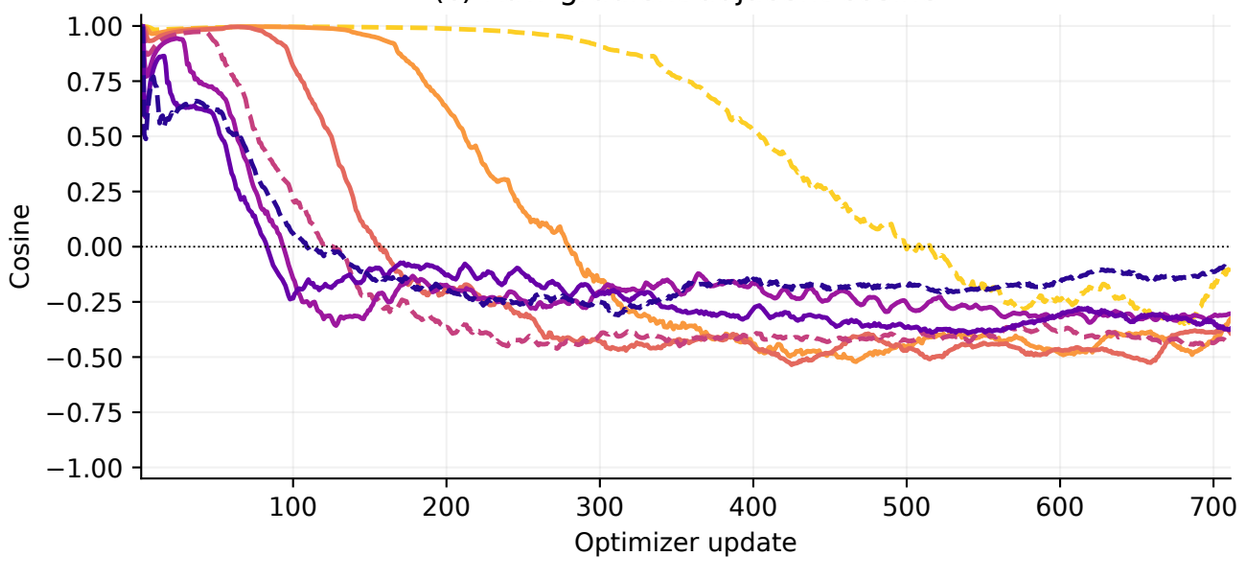
(a) Training loss



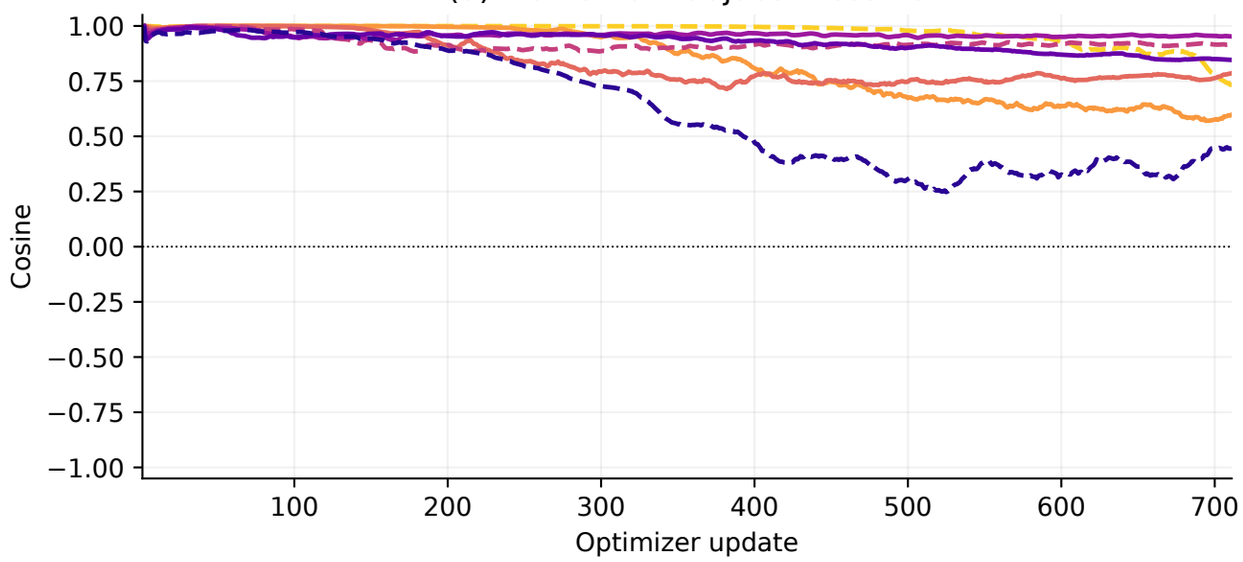
(b) Raw-gradient g: cumulative B/A



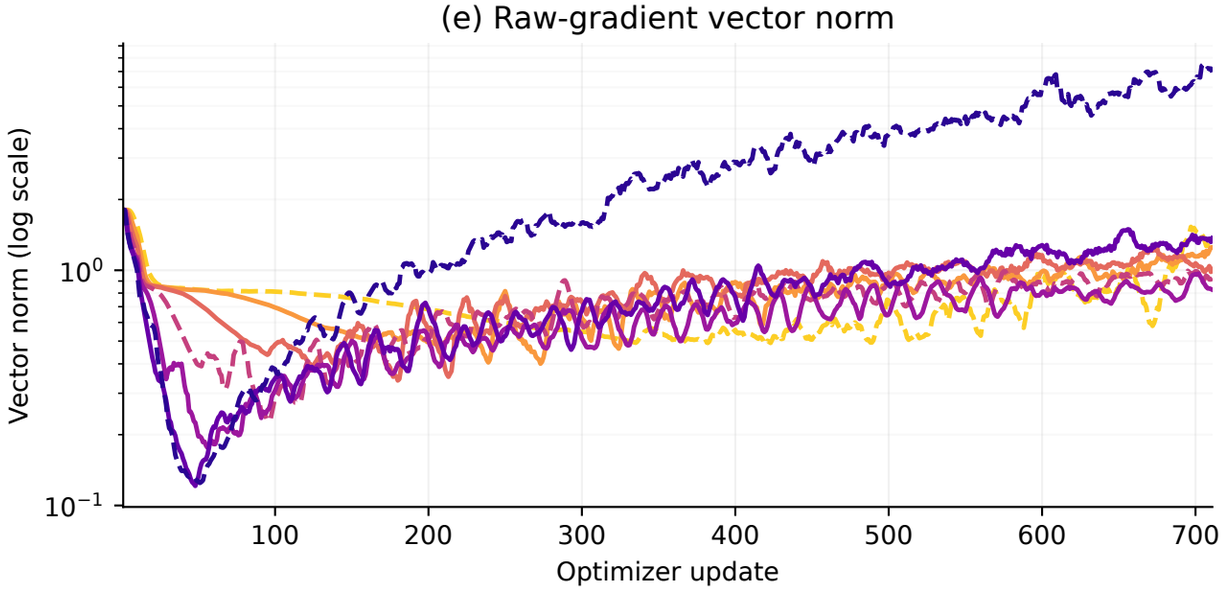
(c) Raw-gradient adjacent cosine



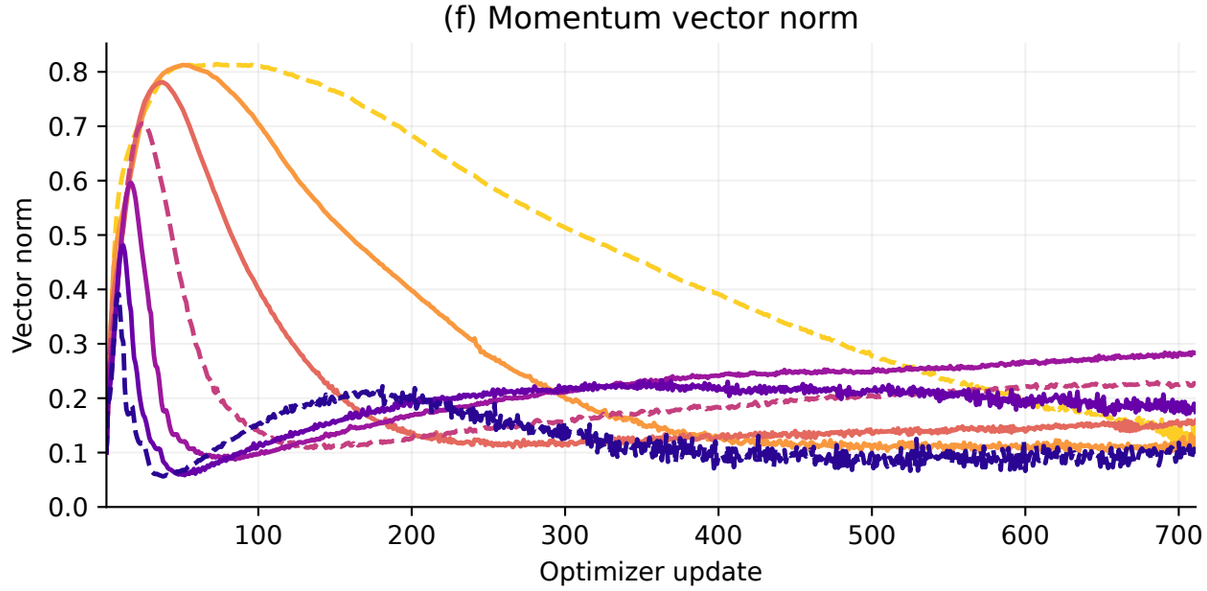
(d) Momentum adjacent cosine



(e) Raw-gradient vector norm



(f) Momentum vector norm



(a) Trailing 25-update token-weighted loss mean; light traces: individual losses. (b) Raw B/A: unsmoothed.
(c) Raw adjacent cosine: trailing 50-update arithmetic mean. (d) Momentum adjacent cosine: trailing 25-update arithmetic mean.
(e) Raw norm: trailing 10-update arithmetic mean, logarithmic Y. (f) Momentum norm: unsmoothed, linear Y.
Windows include the current update, use available prefix data, and exclude undefined first cosines; no future data.
Updates 1-711 shown; partial update 712 retained in results. g is preclip; m is stored exp_avg before preconditioning.
Seed 1234; frozen initialization/order; different hardware across S/R; LR scales AdamW decay; one seed.